

Case Study: Building a LEGO Investment Strategy Through Data Analytics

Introduction

Michael Thompson, a 35-year-old data analyst and father of two young daughters (ages 5 and 7), lives in Austin, Texas. Recently promoted to a senior analyst role at a tech firm, Michael enjoys a stable income and a passion for fostering his children's creativity. While balancing their screen time with traditional toys, he noticed his eldest daughter's growing obsession with LEGO sets. This sparked a personal interest when he read an article, "LEGO Bricks Outshine Gold Bars as Investments, Study Finds" (Burton 2019), suggesting LEGO sets could be more than toys—they could be investments. With his data analytics expertise, Michael aims to explore whether LEGO sets can fund his daughters' hobbies and contribute to their college savings.

Background: LEGO as a Brand and Investment

Founded in 1932 by Ole Kirk Kristiansen in Denmark, The LEGO Group is a global toy leader. Its iconic LEGO brick, patented in 1958, uses a stud-and-tube system ensuring compatibility across decades. From basic sets in five colors, LEGO evolved into complex themes like Star Wars, City, and Technic, appealing to kids and collectors alike.

LEGO sets also hold investment potential. Limited-edition sets can fetch thousands on the secondary market, but even retail sets can appreciate post-retirement—when LEGO ceases production, typically after 6 months to 5 years. For instance, the LEGO Sopwith Camel (Set #10226), which sold for \$99.99 in 2012, averaged \$145 on eBay years later, a 7.71% compound annual growth rate (CAGR). A study by Dobrynskaya and Kishilova (2022) reported an 11% CAGR for LEGO sets from 1987 to 2015, though appreciation varies.

Inspired, Michael wants to leverage data analytics to identify high-value LEGO sets for investment.

The Opportunity

Michael seeks to develop a data-driven strategy to purchase LEGO sets likely to appreciate after retirement. Using a dataset of all 614 LEGO sets released in 2018 and 2019, sourced from Brickset (www.brickset.com), he plans to analyze set characteristics—e.g., theme, piece count, price—to pinpoint "outsized value" opportunities (sets underpriced relative to their potential). His dual goal is to support his daughters' LEGO passion and generate extra savings. The dataset, "LEGO Analytics Case Data.xlsx," excludes promotional or polybag sets and focuses on retail-available sets from late 2019. He will conduct two analyses: descriptive and predictive, using Python for all computations and visualizations.

Descriptive Analysis: Exploring the LEGO Landscape

Michael's first task is to understand the LEGO market by examining the 2018 and 2019 sets across themes and subthemes. He's particularly interested in these investment-relevant themes:

- Architecture, City, Creator, Creator Expert, DC Super Heroes, Disney, Friends, Harry Potter, Hidden Side, Ideas, Jurassic World, LEGO Movie 2, Marvel Super Heroes, Minecraft, Ninjago, Speed Champions, Star Wars, Technic

He also wants to explore two subthemes:

- **Brickheadz:** Blocky, collectible figure statues.
- **Juniors:** Sets for young builders (ages 4–6) with simpler designs.

Michael aims to uncover patterns in set characteristics (e.g., piece count, price, mini-figures) and differences across themes/subthemes. He's open to computed metrics like price per piece or box density (weight/volume) to enrich his insights.

Deliverables for Descriptive Analysis

Using Python, students will:

1. **Summarize Data:** Calculate descriptive statistics (e.g., count, average piece count, average price) for each theme and subtheme, combining 2018 and 2019 data.
2. **Visualize Insights:** Create visualizations (e.g., bar charts for set counts by theme, scatter plots for price vs. piece count) using libraries like Matplotlib or Seaborn.
3. **Submit:**
 - A Python script with code for data loading, processing, and visualization.
 - A report (2–3 pages) summarizing findings, including at least 3 visualizations and interpretations (e.g., which themes have the most sets? Highest price per piece?).
 - Due: 11:59 pm Sun, Apr 27.

Predictive Analysis: Identifying Investment Opportunities

Michael's second task is to predict which 2019 sets offer the best investment value. He'll build a model to estimate what a set "should" cost based on its characteristics, then calculate its **value potential** (predicted price – actual retail price). A set priced at \$29.99 with a predicted price of \$35 has a \$5.01 value potential, suggesting it's underpriced and a potential investment. He'll focus on 2019 sets, as they were still widely available in late 2019, and evaluate:

- Top two and bottom two sets by value potential in each theme, subtheme (Brickheadz, Juniors), and price range (\$19.99–\$29.99, \$34.99–\$69.99, \$74.99–\$99.99, \$100.00+).

Deliverables for Predictive Analysis

Using Python, students will:

1. **Build a Model:** Use a proper regression model to predict retail price, selecting key features (e.g., piece count, weight) from the dataset.
2. **Calculate Value Potential:** Compute value potential for all 2019 sets and identify top/bottom performers as specified.
3. **Submit:**
 - A Python script with code for data preprocessing, model building, and value potential calculation.
 - A report (2–3 pages) detailing the model (e.g., equation, adjusted R^2), results (top/bottom sets with explanations), and investment recommendations for Michael.
 - Due: 11:59 pm Sun, May 04

References

Burton B (2019) LEGO bricks outshine gold bars as investments, study finds. *CNet* (January 28), <https://www.cnet.com/news/lego-bricks-outshine-gold-bars-as-investments-report-says/>.
Dobrynskaya V, Kishilova J (2022) LEGO: The toy of smart investors. *Res. Internet. Bus. Finance* 59:101539.
LEGO (2019) LEGO history website. Accessed April 18, 2022, <https://www.lego.com/en-us/lego-history>.